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Improving the method for calculating carbon emissions from waste incineration: confirmed with carbon-14 testing of flue gas

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Abstract

An improved method was developed to calculate direct emissions from seven municipal solid waste incineration plants by adjusting the physical compositions of waste by invoking the proportion of co-incinerated waste and the bottom ash yield. The fossil carbon fractions of the waste components were determined by carbon-14 (^{14}C) testing. Based on the improved method, direct emissions were in the range of 222–610 kg CO_2-eq/t waste, corresponding to reductions of 3.4–221 kg CO_2-eq/t waste compared with the method without waste composition adjustment. The ^{14}C contents of the flue gas before and after gas cleaning were tested to validate the improved method, and indicated fossil CO_2 emissions of 249–446 and 233–405 kg CO_2-eq/t waste, respectively. The direct emissions obtained by the improved method were closer to the results of ^{14}C testing, due to more accurate estimations of the actual waste composition. The method was further combined with a life cycle analysis of the waste incineration process, obtaining total carbon emissions in the range from −33.2 to 483 kg CO_2-eq/t waste. The findings provide a new means of accurately calculating carbon emissions from waste incineration.

Highlights

- Co-incinerated waste and bottom ash yield were used to adjust waste composition.
- Fossil carbon contents of the waste and flue gas were determined by ^{14}C testing.
- The improved method reduced fossil CO_2 emissions by 3.4–221 kg CO_2-eq/t waste.
- Life cycle GHG emissions from waste incineration was −33.2–483 kg CO_2-eq/t waste.

Keywords Direct emissions, Fossil carbon fraction, Accelerator mass spectrometry, Co-incineration, Bottom ash, Waste composition adjustment

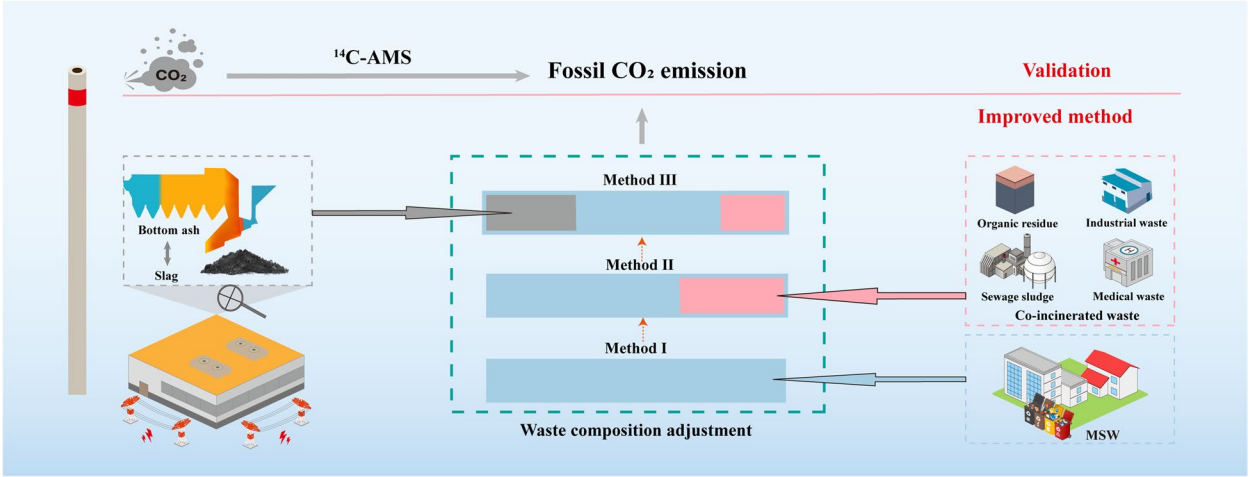
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Graphical Abstract



1 Introduction

Improving the carbon accounting system is a necessary procedure to achieve carbon neutrality (Wu et al. 2022). In the area of solid waste management, municipal solid waste incineration (MSWI) is one of the most promising pathways to reduce greenhouse gas (GHG) emissions, as incineration facilities provide both electricity generation and an alternative to landfilling (Liao et al. 2023; Fei et al. 2024). As the waste composition and energy mix in China have changed over the years, the carbon benefits of waste incineration have incurred a downward trend (China Academy of Environmental Planning 2023; Han et al. 2023; Zhang et al. 2023). Accurate accounting of carbon emissions from MSWI is now widely scrutinized (Zhao et al. 2022; Xiao et al. 2023).

The accounting methods for carbon emissions from MSWI include the method established by the intergovernmental panel on climate change (IPCC) (Eggleston et al. 2006), life cycle assessment (LCA) (Liao et al. 2023), the balance method (Schwarzböck et al. 2016), the carbon-14 (^{14}C) method (Lee et al. 2021), and the clean development mechanism (Yao et al. 2019). Recent studies have primarily focused on the IPCC method and LCA. The IPCC method employs emission factors and formulas to calculate direct emissions from fossil components in waste. The LCA method, which further utilizes operational data from MSWI plants, identifies both indirect emissions from energy and material consumption as well as the offset avoided emissions from electricity generation. Due to abundant operational data and the existence of product life cycle databases, the calculation of indirect emissions and avoided emissions is typically straightforward. However, the calculation of direct emissions is more complex, as determining the fossil carbon fraction (FCF) in waste often involves complicated data research and experimental work. To date, such determination of FCF has involved sorting waste into more uniform components (paper, plastic, glass, metal, textiles, etc.) and analyzing the total carbon content (TC) and the distribution of FCFs among these components (Chen et al. 2020).

Nevertheless, due to a lack of visual identifiability, a significant fraction is often overlooked due to small particle size, with the most significant impact being on the content of inert waste such as dust and debris. In recent studies, the total ash contents of waste components in industrial analysis often appeared to be low (Malakahmad et al. 2017; Pérez et al. 2018; Zhao et al. 2022), but the actual yields of bottom ash in MSWI plants were high (20%–30%) (Sabbas et al. 2003; Li et al. 2020; Chen et al. 2023). This implies that the ash content in the waste was underestimated, while the contents of other components, such as plastics, which have a significant impact on carbon emissions, were overestimated. Furthermore, changes in waste management policies have led to alterations in the reality of MSWI. The implementation of waste sorting policies has resulted in the diversion of a significant amount of food waste from municipal solid waste (MSW) (Shi et al. 2024), leading to a substantial reduction in the waste sent to MSWI plants. Meanwhile, the rapid increase of MSWI plants in recent years has led to a challenge of their underutilization. In response to this situation, MSWI plants have shifted towards the co-incineration of various types of waste, including

non-hazardous industrial solid waste, disinfected medical waste, dried sludge, residues from food waste treatment, and so on (Zhao et al. 2021; Ying et al. 2023; Guo et al. 2023; Chu et al. 2024). However, the components of these co-incinerated wastes are typically not considered in MSWI carbon emission accounting (Zhao et al. 2022; Liao et al. 2023; Han et al. 2023).

Currently, research efforts on the FCFs of various waste components primarily rely on the default values provided by IPCC guidelines (Kourkoumpas et al. 2015; Liu et al. 2017; Wang et al. 2017), which may not accurately reflect the characteristics of China's waste. Studies have indicated differences between the IPCC default values and the actual FCFs (Kang et al. 2017a; Liu et al. 2024). Therefore, to improve the reliability of accounting results, it is necessary to determine the FCF of each component in waste. The ^{14}C test is an effective method for quantifying the fossil carbon content in waste. Theoretically, modern biomass has 100% radiocarbon content, in contrast to fossil fuels, which contain virtually no radiocarbons (American Society for Testing and Materials 2024). The biogenic fraction correlates directly with the ^{14}C content, allowing for its precise determination. Consequently, the ^{14}C method adeptly differentiates between fossil and biogenic carbon sources (Kang et al. 2020). This approach ensures the accurate assessment of fossil carbon in waste. Additionally, employing ^{14}C testing to determine the concentrations of fossil CO_2 in flue gas is a more precise method for directly calculating carbon emissions from waste incineration (Wang et al. 2022). The flue gas from incineration is the most suitable target for validating direct carbon emissions (Kim et al. 2016; Kang et al. 2018), because the homogeneous nature of flue gas allows for avoidance of the issue raised by unrepresentative sampling of heterogeneous solid waste (Mohn et al. 2012). Nevertheless, the challenges of flue gas sampling and the expense of ^{14}C testing often discourage researchers from conducting extensive sampling and tests.

Therefore, this study proposes an improved method for determining direct emissions from MSWI plants by invoking the adjusted physical composition of waste. The improvement comprises two steps: step 1, adjusting the physical composition of incinerated waste based on the quantities of co-incinerated waste materials; step 2, adjusting the physical composition from step 1 based on the actual bottom ash production. Additionally, ^{14}C testing was adopted to determine the FCF of each component in waste, providing carbon emission accounting parameters that align with the current situation in China, further improving the method. To validate the results, direct emissions from waste incineration under eight different conditions were determined through ^{14}C testing of flue gas. The direct emissions determined by the improved method were then combined with a full life cycle inventory analysis to evaluate the current level of carbon emissions from MSWI in China.

2 Method

2.1 MSWI plants

Seven typical MSWI plants in Shanghai (plants A–E), Jiangsu (plant F), and Hebei province (plant G) in China were selected as study cases, with eight sets of operating conditions. For plant G, two sets of working conditions were operative, namely incineration of MSW alone, denoted as G(1), and co-incineration of MSW with 10% polyester textiles, denoted as G(2). The three regions selected, with different climatic and economic conditions, level of development and population densities (179,907, 144,390 and 56,995 CNY for per capita gross regional product, and 3950, 794 and 393 persons/km^2 for population density in 2022 in Shanghai, Jiangsu, and Hebei province, respectively), ensured that the waste samples were broadly representative. The waste management practices in the three regions, in turn, represent the typical waste segregation patterns and levels today. At the MSWI plants A–E sites, MSW was collected separately as hazardous waste, recyclables, food waste and residual waste, with residual waste ultimately being sent to MSWI plants. At the locations of MSWI plant F and MSWI plant G, MSW was collected separately as hazardous waste, recyclables and other waste, so the residual waste sent to the MSWI plant had a high food waste content. The incineration sizes of the MSWI plants in the three regions ranged from 750 t/d to 6,000 t/d, and the incineration and pollutant management technologies used covered typical waste incineration scenarios. During the period from November 2022 to November 2023, samples and operational data were collected from the MSWI plants. The main parameters and system indicators of the investigated plants are shown in Table 1.

Table 1 Characteristics of the investigated MSWI plants

Facility	Incinerator type	Capability (t/day)	Live steam pressure/temperature (MPa/℃)
Plant A	Grate incinerator	1000	4.0/400
Plant B	Grate incinerator	1500	4.0/400
Plant C	Grate incinerator	3500	6.4/450
Plant D	Grate incinerator	6000	5.4/450
Plant E	Grate incinerator	1500	4.0/400
Plant F	Grate incinerator	750	4.0/400
Plant G(1)	Grate incinerator	2000	13.5/450
Plant G(2)	Grate incinerator	2000	13.5/450

2.2 Sample collection and characterization

Besides MSW, other types of waste are co-incinerated in the plants. Therefore, sample collection and analysis of the physical compositions of incoming waste materials at the plants became more complicated, covering MSW as well as blended industrial solid waste, medical waste, sludge, and food waste treatment residues. MSW, as the bulk of the incoming waste, was sampled in the same way as in the study by Liao et al. (2023). The amounts of other co-incinerated waste materials were obtained from the operational data. Typical co-incinerated waste components included polyester textiles and rubber in industrial solid waste; syringes, surgical drape, surgical gowns, and medical pads in medical waste; organic solid residues and biogas residues in food waste treatment residues; and sludge (Table 2).

Each typical component of the MSW and co-incinerated waste materials was dried and crushed separately, and then passed through a 100-mesh sieve to ensure homogeneity of the samples. The TC of the samples was determined by elemental analysis (Vario EL Cube, Elementar, Germany). The FCFs of the samples were determined by accelerator mass spectrometry (AMS). Each solid waste sample with an absolute carbon content of 200–1000 μg was placed in a quartz tube to which an excess of copper oxide particles had been added, which was then connected to a vacuum system. The quartz tube containing the sample was then sealed and placed in a muffle furnace at 850 °C for 1 h, during which the organic carbon was oxidized to CO_2. The gaseous products were then collected for ^{14}C testing, the protocol for which is described in Sect. 2.4.

2.3 Flue gas sampling and CO_2 analysis

The method ASTM D7459-08 (American Society for Testing and Materials 2016) defines specific procedures for the collection of gas samples from stationary emission sources for subsequent laboratory determination of the ratio of fossil carbon to biogenic carbon according to method ASTM D6866-22 (American Society for Testing and Materials 2024). In this study, a sampling method was established with reference to ASTM D7459-08, and flue gas samples were collected from seven MSWI plants operated under eight sets of conditions. Two sampling points were selected at each MSWI plant, namely the boiler economizer outlet and the chimney outlet, to determine the total CO_2 and fossil CO_2 concentrations in flue gas before and after the gas cleaning system, respectively.

Figure 1 shows a schematic diagram of the flue gas collection and monitoring. A primary filter made of stainless steel was set up at the back end of the 1 m long sampling probe to prioritize the removal of particles larger than 0.1 mm in the flue gas. The flue gas was then passed through a tracer line maintained at 180 °C to avoid losses of its components due to condensation. A specific condensation system (PSS-20, M&C TechGroup, Germany), equipped with a polytetrafluoroethylene gas filter to remove particles larger than 2 μm, was used to remove moisture from the flue gas. The condensation system

Table 2 FCFs in this study and IPCC default values

IPCC category	FCF category in this study	FCF (%)		TC value in this study (dry basis)
		Average in this study	IPCC default value	
Food waste	Food waste	3.67 ± 0.95	0	41.74 ± 1.69
Paper and cardboard	Paper	3.51 ± 0.69	1	40.53 ± 2.56
Plastics	Plastics	90.52 ± 0.81	100	74.01 ± 6.20
	Syringe	89.28 ± 1.05		49.63 ± 0.19
Textiles	Surgical drape	98.71 ± 0.08	20	76.29 ± 0.66
	Surgical gown	98.71 ± 0.01		73.35 ± 0.78
	Cotton textiles	0.00		52.54 ± 8.45
	Polyester textiles	93.60 ± 0.02		45.64 ± 1.14
	Mixed textiles	62.40 ± 0.01		47.94 ± 2.92
Wood	Wood	0.64 ± 0.46	0	47.07 ± 0.78
Nappies	Nappies	73.11 ± 7.28	10	56.78 ± 7.30
	Medical Pads	74.06 ± 0.30		46.28 ± 3.38
Rubber and leather	Rubber	94.98 ± 2.24	20	86.19 ± 0.01
No corresponding category	Biogas residue	1.26 ± 0.09	/	23.98 ± 0.47
	Organic solid residue	3.09 ± 1.92		44.06 ± 2.51
	Sludge	23.08 ± 0.59		27.90 ± 0.01

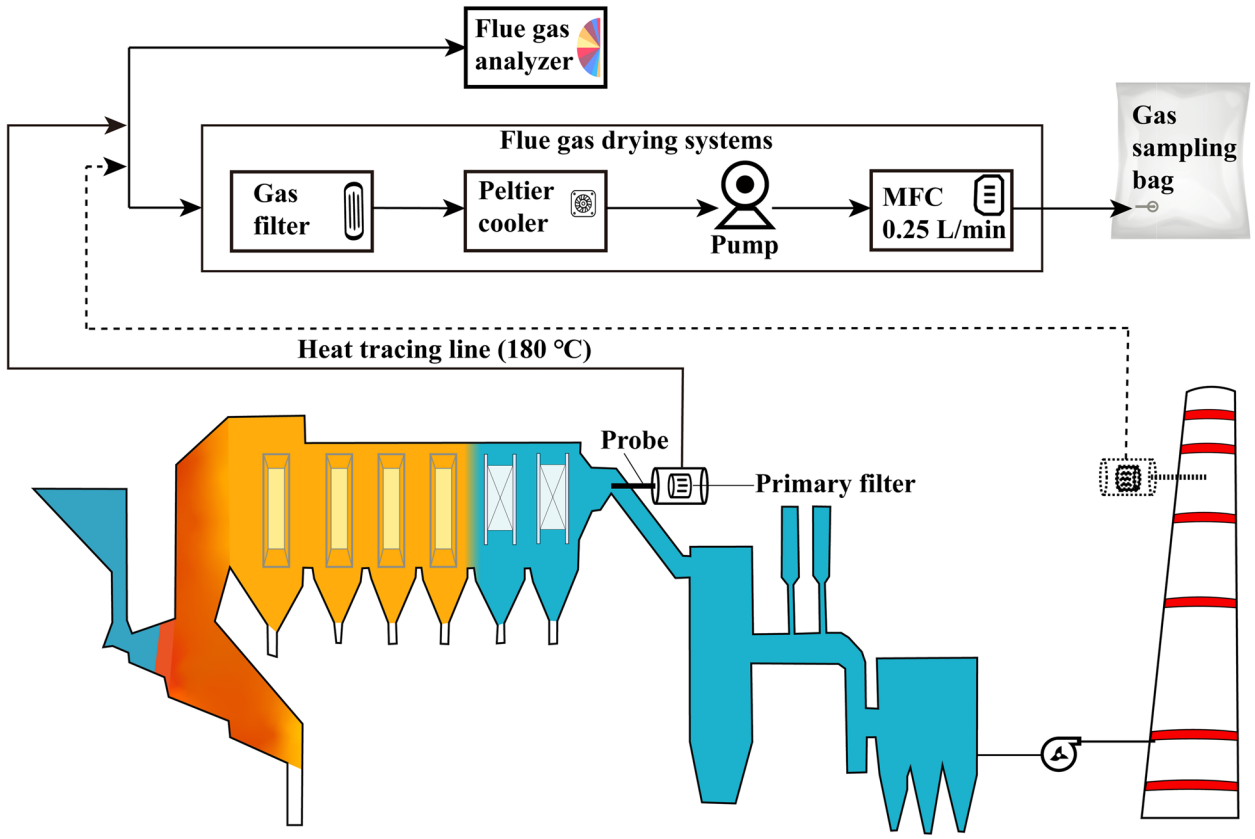


Fig. 1 Schematic diagram of the field setup for flue gas sampling. MFC: mass flow controller

featured a high-performance Peltier electronic condenser for rapid condensation (below 5 °C) and dehydration of the flue gas. A gas pump provided power to the sampling system, and a mass flow controller (MFC) ensured that the flue gas was collected in a Devex gas sampling bag at a flow rate of 0.25 L/min during a continuous sampling period of 1 h.

CO_2 concentrations in the flue gas were tested using a gas analyzer (DX4000, Gasmet Technologies Oy, Vantaa, Finland). The flow rates of flue gas were determined according to the continuous emission monitoring systems of the respective MSWI plants. It is worth noting that, due to the difficulty of obtaining flue gas flow rate data at the boiler, this study chose to overlook the change in flue gas flow rate after the gas cleaning system. It is assumed that the flue gas flow rate at the outlet of the boiler economizer is consistent with that at the chimney outlet, and the latter was used for the calculation (Bellani et al. 2023).

2.4 ^{14}C test

Flue gas samples and gaseous products from solid waste components were purified and graphitized prior to ^{14}C testing by AMS. Gas samples were transferred to a vacuum system at a flow rate of 200 mL/min, in which H_2O and CO_2 were sequentially frozen by passage through a liquid nitrogen cold trap (−196 ℃), while other impurity gases were pumped out of the system through a vacuum line. The H_2O was then removed by an ethanol-liquid nitrogen water-cooling trap (−78 ℃ to −90 ℃) to obtain purified CO_2. The purified CO_2 was then transferred to a glass reaction casing, the outer tube of which was filled with Zn and TiH_2 powders, and the inner tube was filled with Fe powder. After flame-sealing under vacuum, the glass casing was heated in a muffle furnace. CO_2 underwent a reduction reaction on the surface of the Fe powder in the inner tube after successive stepwise heating at 500 °C for 3 h and at 550 °C for 4 h, and was finally reduced to graphite. The graphite sample was then pressed into an aluminum support prior to analysis.

The ^{14}C levels of the graphite samples were measured by AMS (0.5 MV, NEC, USA) and the errors in the ^{14}C measurements were within 3‰ of the ^{14}C concentration of the samples. The biological carbon fractions (BCFs) of the samples were expressed in terms of modern carbon fraction $f_{M,sample}$ (unit: pMC). The test data were

calibrated externally using oxalate (OXII) (f_M = 134.07 pMC) as a primary standard and hyaluronan (HA) (f_M = 65.76 pMC) as a secondary standard. The isotopic mass fractionation index correction model was used to correct $^{14}C/^{13}C$ using $\delta^{13}C$ data from the AMS online test, as expressed by Eq. (1):

$$f_{M,sample} = \frac{(^{14}C/^{13}C)_{sample}}{(^{14}C/^{13}C)_{ABS}} \quad (1)$$

where $(^{14}C/^{13}C)_{sample}$ is the $^{14}C/^{13}C$ ratio of the sample normalized to the $\delta^{13}C$ value, and $(^{14}C/^{13}C)_{ABS}$ is the absolute radiocarbon reference standard after the application of fractionation and decay corrections.

It was necessary to adjust all $f_{M,sample}$ values by applying an atmospheric reference correction factor (REF) to obtain the true BCFs of the samples. According to ASTM D6866-22, modern atmospheric levels of ^{14}C are constantly being diluted as fossil fuel combustion continues to emit fossil carbon into the modern atmosphere, currently with an atmospheric $f_{M,REF}$ of 100.00 pMC. BCF and FCF can be calculated according to Eq. (2):

$$\mathrm{BCF} = 100\% - FCF = \frac{f_{M,sample}}{f_{M,REF}} \quad (2)$$

2.5 Fossil CO_2 emissions accounting

2.5.1 Method I based on MSW physical composition

Combining specific physical composition data of waste from different MSWI plants, the fossil CO_2 emission factor (FCO_2EF) of MSW was determined according to Eq. (3) (Han et al. 2023; Ma et al. 2024).

$$FCO_2EF(1) = \sum_{i=1}^{n}\left(\frac{WF_i}{100}\times\frac{DM_i}{100}\times\frac{CF_i}{100}\times\frac{FCF_i}{100}\right)\times\frac{44}{12}\times 1000 \quad (3)$$

where $FCO_2EF(1)$ denotes the fossil CO_2 emission from waste incineration calculated by Method I, kg CO_2-eq/t waste; WF_i is the weight fraction of component i in the MSW on a wet basis (%); DM_i is the dry matter (DM) content of component i by weight (%); CF_i is the weight fraction of TC in the DM of component i (%); FCF_i is the weight fraction of fossil carbon in the TC of component i (%); 44/12 is a conversion factor relating C to CO_2; and i is a given component in the MSW, $i = 1, 2, \ldots, n$.

2.5.2 Method II improved by invoking co-incinerated waste materials

Through data recorded from the MSWI plants, the other co-incinerated waste materials were included in the physical composition of the incinerated waste in each case. The calculation of carbon emissions from waste incineration was improved by applying Eq. (4) based on these co-incinerated waste materials.

$$FCO_2EF(2) = FCO_2EF(1)\times R_1 + \sum_{j=1}^{m}\left(\frac{WF_j}{100}\times\frac{DM_j}{100}\times\frac{CF_j}{100}\times\frac{FCF_j}{100}\right)\times\frac{44}{12}\times 1000 \quad (4)$$

where $FCO_2EF(2)$ denotes the fossil CO_2 emission from waste incineration calculated by Method II, kg CO_2-eq/t waste; R_1 is the proportion of MSW in the incinerated waste; WF_j is the weight fraction of the co-incinerated waste j in the total incinerated waste on a wet basis (%); DM_j is the DM content of waste j by weight (%); CF_j is the weight fraction of TC in the DM of waste j (%); FCF_j is the weight fraction of fossil carbon in the TC of waste j (%); and j is the type of co-incinerated waste, $j = 1, 2, \ldots, m$.

2.5.3 Method III modified by invoking co-incinerated waste materials and bottom ash

Further adjustment of the physical compositions of incinerated waste samples was based on the actual bottom ash yields from the MSWI plants, as expressed by Eqs. (5) and (6):

$$R_2 = \frac{\frac{BA}{100}\times\frac{DM_{BA}}{100}\times\frac{(100-P_{BA})}{100} - R_1\times\sum_i^m\left(\frac{WF_i}{100}\times\frac{DM_i}{100}\times\frac{Ash_i}{100}\right) - \sum_j^m\left(\frac{\mathrm{WF_j}}{100}\times\frac{\mathrm{DM_j}}{100}\times\frac{\mathrm{Ash_j}}{100}\right)}{1 - \frac{M_{inert}}{100} - \left[R_1\times\sum_i^m\left(\frac{WF_i}{100}\times\frac{DM_i}{100}\times\frac{Ash_i}{100}\right) + \sum_j^m\left(\frac{\mathrm{WF_j}}{100}\times\frac{\mathrm{DM_j}}{100}\times\frac{\mathrm{Ash_j}}{100}\right)\right]} \quad (5)$$

$$FCO_2EF(3) = FCO_2EF(2)\times(1 - R_2) \quad (6)$$

where R_2 is the percentage of neglected inert components in the incinerated waste; BA is the bottom ash yield in the incineration plant (%); DM_{BA} is the DM content of bottom ash by weight (%); P_{BA} is the combustible fraction of bottom ash (%); Ash_i is the ash content of MSW component i on a dry basis (%); Ash_j is the ash content of co-incinerated waste j on a dry basis (%); M_{inert} is the water content of neglected inert components in the incinerated waste, which is supposed to be the same as the fraction of dust and debris detected (25.0%); and $FCO_2EF(3)$ is the fossil CO_2 emission from waste incineration calculated by Method III, kg CO_2-eq/t waste.

2.6 Method validation based on analyzed fossil CO_2 concentrations in flue gas

Fossil CO_2 emissions were determined by combining the flow rate, CO_2 concentration, and the FCF_{fg} of flue gas. Method IV and Method V determine the fossil CO_2 emissions before and after flue gas treatment by using untreated flue gas and treated flue gas parameters, respectively, according to Eq. (7):

$$FCO_2EF = \left(Q_n \times \frac{C_{CO2}}{100} \times \frac{44}{22.4}\right) \times \frac{FCF_{fg}}{100} \quad (7)$$

where Q_n is the volumetric flow rate of dry flue gas at the stack outlet under standard conditions (273.15 K, 101,325 Pa) (m^3/t waste); C_{CO2} is the CO_2 concentration in dry flue gas under standard conditions (273.15 K, 101,325 Pa) (volume%); and FCF_{fg} is the fossil CO_2 fraction in the total CO_2 (%).

2.7 Total GHG emissions associated with waste incineration

When the LCA method is used, the total GHG emission (GHG_{total}) from waste incineration includes the direct emission (GHG_{direct}) due to fossil CO_2 emission, indirect emission ($GHG_{indirect}$), and avoided emission (GHG_{avoid}), calculated according to Eq. (8).

$$GHG_{total} = GHG_{direct} + GHG_{indirect} + GHG_{avoid} \quad (8)$$

Indirect emissions in MSWI plants are associated with auxiliary fuel and electricity consumption, flue gas cleaning, leachate treatment, and fly ash/bottom ash treatment and disposal. $GHG_{indirect}$ is determined by multiplying the consumption of energy and materials involved in the indirect emissions process with the relevant product carbon footprint factors. The MSWI plants studied here were located in three different geographic locations and had varying waste collection and transportation routes. Considering that emissions from the collection and transportation phase are extremely low compared to the overall carbon emission level of waste incineration (Pfadt-Trilling et al. 2021; Zhao et al. 2022), they were excluded from the indirect emission accounting in this study. Avoided emissions from waste incineration plants were embodied in energy recovery in terms of electricity, determined by multiplying the amount of electricity generated from waste incineration by the carbon emission factor of the electricity in which the MSWI plant is located. Due to variations in electricity generation efficiency and regional carbon emission factors for electricity, different MSWI plants achieved varying degrees of GHG emission reduction. The life cycle inventory of indirect and avoided emissions is shown in Table S1. The calculation was conducted using Microsoft Excel, and the result was discounted to the carbon emissions from incineration of one tonne waste. For the calculation, the product carbon footprint factor data were obtained from the Ecoinvent database and the China Product Carbon Footprint Factor Database. The actual operational data used were divided into three parts according to the type of GHG emissions, where the operational data for indirect and avoided emissions were collected from the incineration plant ledger, and the data related to direct emissions were obtained by investigating the physical and chemical properties of the waste.

3 Results and discussion

3.1 Influence of FCF on fossil CO_2 emissions by Method I

The FCFs of the MSW components obtained by ^{14}C testing showed characteristic differences from the IPCC default values (Table 2). As a result, the direct carbon emissions from waste incineration calculated according to Method I using these two sets of data also showed some differences (Fig. 2).

Plastic waste is the main contributor to fossil CO_2 emissions from MSW due to its high proportion therein and high FCF. The analyzed FCF of plastic in MSW was 90.52% ± 0.81%, somewhat lower than the IPCC default value of 100%, leading to a significant difference in the fossil CO_2 emissions from plastic incineration (Fig. 2). For textiles, significant differences in FCF between cotton and polyester textiles were found, 0% and 93.6% ± 0.02%, respectively, whereas the IPCC stipulated an average value of 20% based on surveys and expert opinion. In order to provide an FCF applicable to textiles in Chinese MSW, a value of 62.4% ± 0.01% was calculated for mixed textiles therein, based on the ratio of pure natural cotton textiles to polyester textiles produced in China (1:2) (China Association of Circular Economy 2023). In the case of a higher proportion of mixed textiles in MSW (plants A–E and plant F), the fossil CO_2 emissions from textiles evaluated using the analyzed FCF and IPCC default values were significantly different. In plant G(2), in which 10% polyester textiles are co-incinerated, the fossil CO_2 emission from the polyester textiles fraction using the analyzed FCF (157 kg CO_2-eq/t waste) was 123 kg CO_2-eq/t waste higher than that using the IPCC default, making it the main carbon emission contributor. The analyzed FCF results for food waste (3.67% ± 0.95%), paper (3.51% ± 0.69%), and wood fractions (0.64% ± 0.46%) were slightly higher than the IPCC defaults, and the analyzed FCF for nappies (73.11% ± 7.28%) was significantly higher than the IPCC default. However, due to the low FCF values of the first three fractions and the low proportion of nappies in MSW, they did not lead to a significant increase in fossil CO_2 emissions.

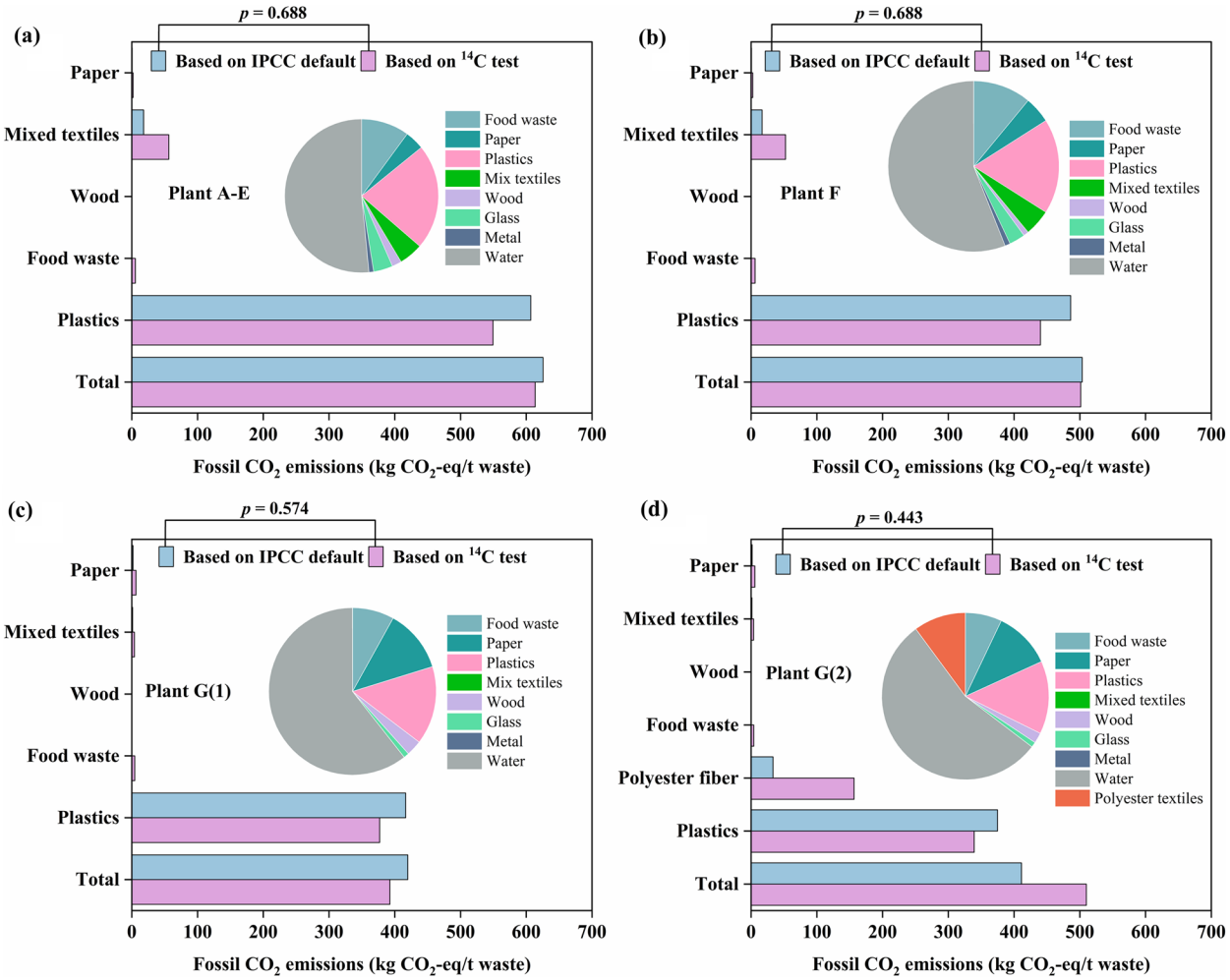


Fig. 2 Physical composition of MSW and fossil CO_2 emissions. The t-test and Mann–Whitney test were used to compare the fossil CO_2 emissions from calculated by different methods, and the significant levels were expressed as P values

Overall, there was no significant difference in fossil CO_2 emissions between the ^{14}C test values and the IPCC default values (Fig. 2), but when the proportions of plastics and textiles were high, the difference in fossil CO_2 emissions between the two methods reached a maximum of 98.73 kg CO_2-eq/t waste. This implies that it is necessary to use the tested FCF data for waste components to accurately calculate the fossil CO_2 emissions from MSWI plants in various regions of China with large differences in the physical composition of waste.

Considering the co-incineration in MSWI plants, FCF test data of other typical components were determined, including industrial waste (rubber), medical waste (syringes, surgical drapes, surgical gowns, and medical pads), food waste treatment residues (biogas residue, and organic solid residue), and sludge (Table 2). These were used to determine parameters commensurate with the characteristics of China's solid waste, which should expedite carbon emission accounting at MSWI plants.

3.2 Fossil CO_2 emissions accounting by improved methods

3.2.1 Improvement by consideration of co-incinerated waste materials (Method II)

Figure 3a presents the physical compositions of waste and fossil CO_2 emissions of MSWI plants A–G calculated according to Methods I and II with analyzed FCF values. Obviously, co-incineration reduces the proportion of MSW in the incinerated waste, such that the relative content of plastic waste, which has a significant impact on fossil CO_2 emissions (Shen et al. 2020), also decreases. Except for plants B, G(1), and G(2), which did not co-incinerate other types of waste, in the other five MSWI plants, the plastic waste contents (dry basis) were reduced from 17.91%–22.36% (Method I) to 17.27%–20.23%

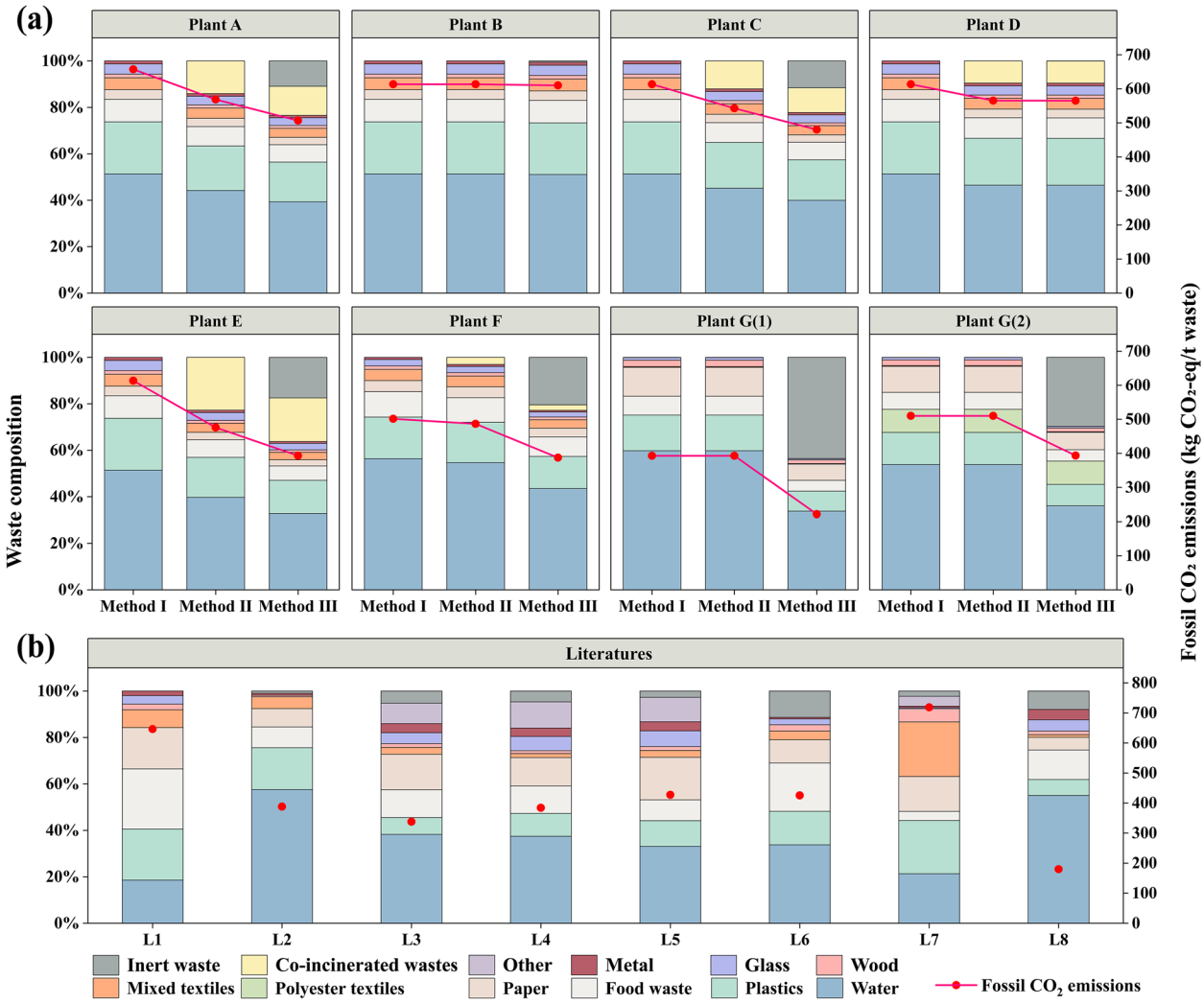


Fig. 3 Results of physical composition of incinerated waste and fossil CO_2 emissions under different accounting methods and in various literature reports (Kourkoumpas et al. 2015; Liu et al. 2017; Wang et al. 2017; Malakahmad et al. 2017; Pérez et al. 2018; Zhao et al. 2022)

(Method II) due to co-incineration. Correspondingly, fossil CO_2 emissions from plastic waste incineration decreased from 501–613 kg CO_2-eq/t waste to 476–565 kg CO_2-eq/t waste. Compared to the commonly used Method I, the total fossil CO_2 emissions of the five MSWI plants using Method II, which considers the proportion of co-incinerated waste, were reduced by 2.95%–22.44%. The proportions and types of co-incinerated waste materials are crucial factors influencing fossil CO_2 emissions. In plant E, due to the co-incineration of 23% garden waste, the content of plastic waste was reduced without introducing other fossil source waste materials. This led to a decrease in fossil CO_2 emissions from 613 kg CO_2-eq/t waste to 476 kg CO_2-eq/t waste, making it the most pronounced case of reducing fossil CO_2 emissions through co-incineration of biogenic waste. In fact, through investigation of co-incinerated waste components (Table S2), it was found that the waste materials currently co-incinerated in MSWI plants mainly include residues from food waste treatment, dried sludge, and paper, which have relatively high BCFs. This significantly reduces the FCFs in the incinerated waste. Therefore, Method II, which considers the proportions of co-incinerated waste materials, can further improve the accuracy of fossil CO_2 emissions accounting in MSWI plants.

3.2.2 Improvement by considering co-incinerated waste materials and bottom ash (Method III)

Ash residues were analyzed for each physical component of the incinerated waste samples. The theoretical bottom ash yield of each MSWI plant was calculated based on the ash contents and physical composition data. Actual

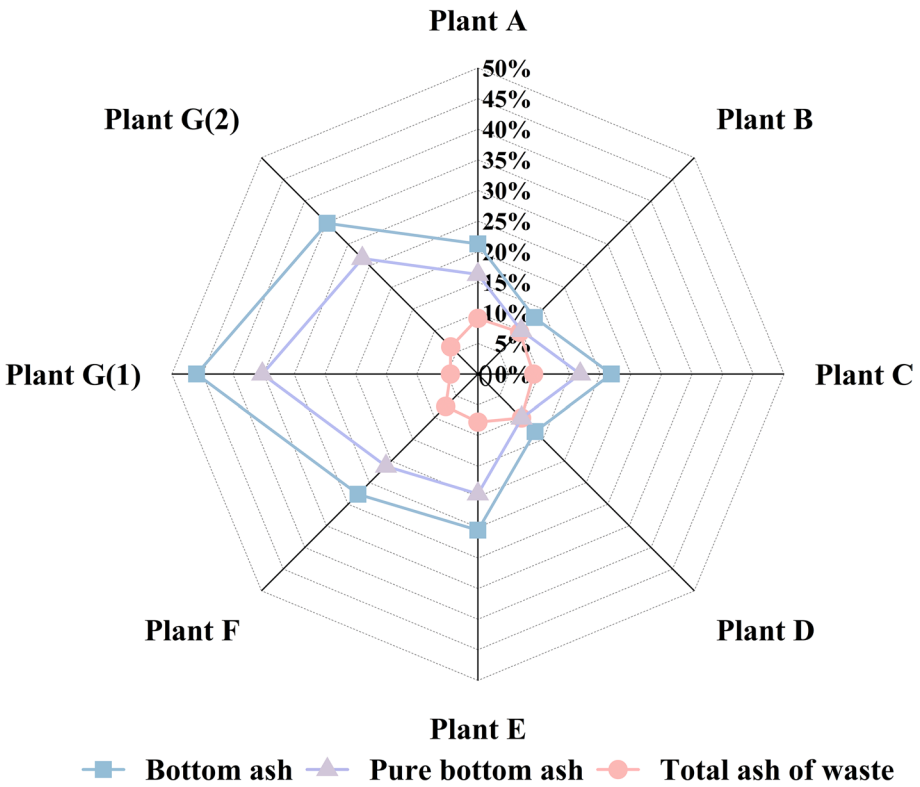


Fig. 4 Total ash contents of incinerated waste versus actual bottom ash yields

bottom ash yield data for each MSWI plant were then compared and differences were analyzed (Fig. 4).

The total ash contents from the incinerated waste materials ranged from 4.50% to 10.09%, while the actual bottom ash yields (wet weight) accounted for 13.08%–45.95% of the waste. Because the bottom ash from MSWI plants was subjected to water quenching before collection and weighing, it had a certain moisture content (Tang et al. 2015; Shen et al. 2021). After deducting the contents of moisture and combustible materials, pure bottom ash yields were in the range 10.02%–35.21%. A significant discrepancy between the total ash residues of incinerated waste and the pure bottom ash yield is thus apparent. Considering the sampling and analysis process for determining physical composition, this can be mainly attributed to the neglect of inorganic particles due to waste heterogenicity and a lack of visual recognizability.

In Fig. 3a, Method III shows the contents of neglected inorganic fractions in incinerated waste (0.04%–43.56%) for MSWI plants A–G using actual pure bottom ash yields. The physical compositions of the waste materials were then re-determined based on the content of the inorganic fraction. It is evident that the consideration of neglected inorganic fractions, as adjusted by Method III, resulted in different percentage reductions in other waste fractions for different MSWI plants. This confirms the necessity of using bottom ash yield to adjust the physical compositions of waste materials from different MSWI plants. Decreases in the proportion of plastic waste were found, by 0.01%–6.69% compared to Method II and by 0.12%–8.10% compared to Method I. Since the inorganic fractions do not participate in incineration or contribute to fossil-derived CO_2 emissions, the fossil CO_2 emissions further decreased to 222–610 kg CO_2-eq/t waste using Method III. This equates to reductions of 0.2–171 kg CO_2-eq/t waste compared to Method II and 3.4–221 kg CO_2-eq/t waste compared to Method I.

In previous literature reports (Fig. 3b), the plastic fraction contents ranged from 6.92% to 22.93%, and the fossil CO_2 emissions ranged from 180 to 719 kg CO_2-eq/t waste. This range covers the results under Methods I–III. However, the non-combustible fraction was less than 5% in most of the studies, reaching 7.90% and 11.30% in only two cases. The non-combustible content in waste is significantly influenced by the subjective judgment of the sampling analysts. Further validation and adjustment may be based on the actual input and output conditions of incineration. The use of actual bottom ash yield in

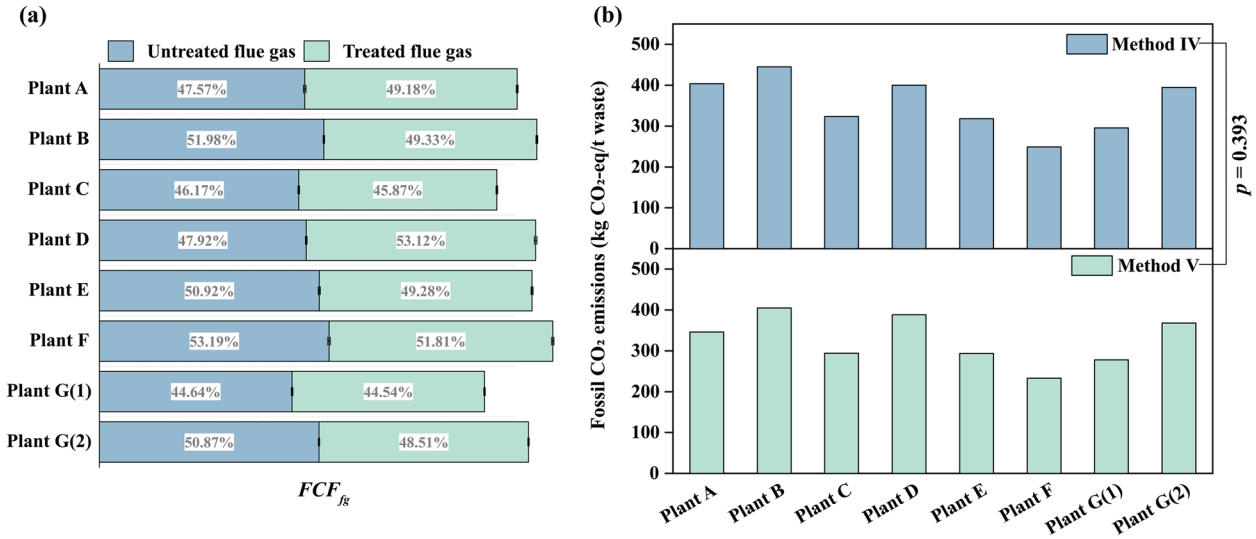


Fig. 5 FCF_{fg} and fossil CO_2 emissions from flue gas. The t-test was used to compare the fossil CO_2 emissions calculated by different methods, and the significant levels were expressed as *P* values

incineration plants may assist in more accurately determining the physical composition of waste and fossil CO_2 emissions under different incineration conditions.

Due to differences in the characteristics of the incinerated waste, the variation in the physical composition of the waste substantially differed among various MSWI plants when applying Method II and Method III for adjustments and calculations. MSWI plants B and G(1) did not co-incinerate other waste materials, leading to least impact on their physical composition when utilizing Method II. In contrast, other plants that accepted an additional 3.04% to 22.75% of other waste types experienced varying degrees of reduction in the components of MSW. The composition of inert waste, such as dust and debris, in the incoming waste varied significantly across different MSWI plants. Method III considered these inert components in the waste composition, further decreasing the proportions of other MSW components in the overall incinerated waste. Similarly, due to variations in waste characteristics, the sensitivity of different incineration plants to Method III differed considerably. Compared to Method II, Method III indicated a reduction of 0.03% to 43.56% in the total content of MSW components (excluding the supplementary inert components) across different incineration plants.

3.3 Validation of the improved method based on flue gas analysis

The flue gas samples collected from MSWI plants A–G were subjected to ^{14}C testing to determine the proportion of fossil CO_2 in the total CO_2. Successful development and implementation of the ^{14}C method for measuring FCF_{fg} from MSWI plants was demonstrated by Mohn et al. (2008). Validation was performed by the balance method (Fellner et al. 2007), as a completely independent method, at three different MSWI plants, and the results of both methods were in good agreement. Furthermore, the ^{14}C test method for flue gas has been successfully applied for the measurement of CO_2 emissions from the incineration of sludge (Kang et al. 2017b), industrial waste (Kang et al. 2020), and solid refuse fuel (Lee et al. 2021).

In Fig. 5a, the FCF_{fg} (proportion of fossil CO_2 in the total CO_2) in untreated flue gas and treated flue gas are seen to be 44.64%–53.19% and 44.54%–53.12%, respectively, in general agreement with the test results of (Mohn et al. 2012). and (Palstra and Meijer 2010). The CO_2 concentrations in the flue gas and other data in Table S3 were used to evaluate emissions of fossil CO_2 from untreated flue gas (Method IV) and treated flue gas (Method V), which resulted in ranges of 249–445 kg CO_2-eq/t waste and 233–405 kg CO_2-eq/t waste, respectively (Fig. 5b). The t-test showed that there was no significant difference in the results of fossil CO_2 emissions determined by Method IV and Method V ($p = 0.393$). The flue gas cleaning systems of the MSWI plants did not significantly change the relative proportions of fossil CO_2 in the total CO_2, but they affected the total amount of CO_2 in the flue gas. After treatment, the CO_2 concentrations in the flue gases of MSWI plants A–G were reduced by 0.26%–2.14%. This was due to CO_2 fixation by $Ca(OH)_2$ and other alkaline substances, the dissolution of CO_2 in water, and the dilution of CO_2 by air infiltration in the flue gas system.

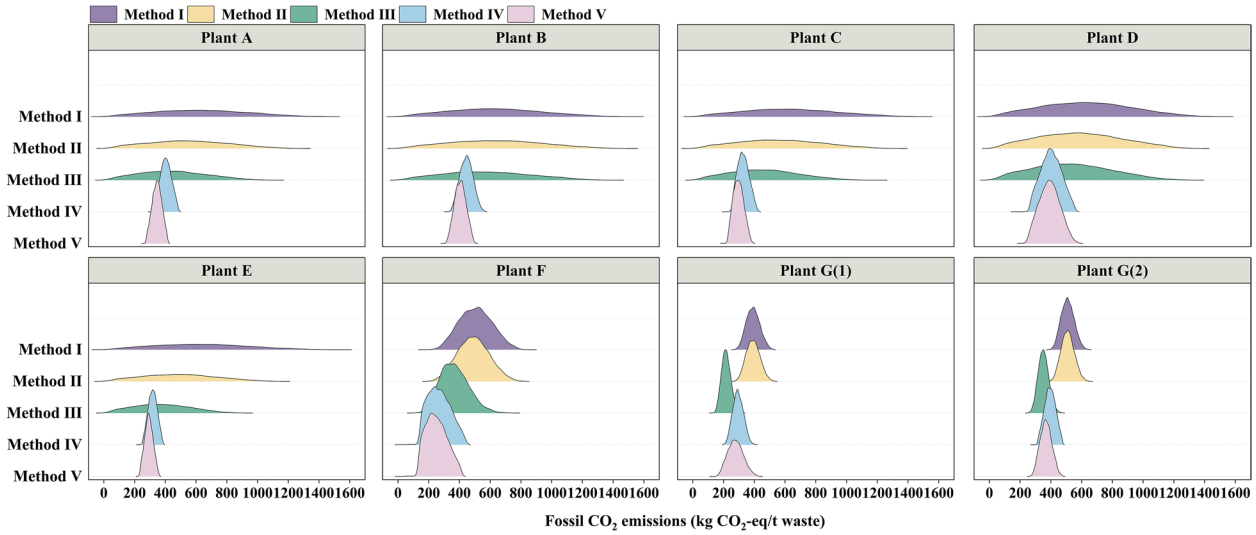


Fig. 6 Distribution of calculation results for the five methods

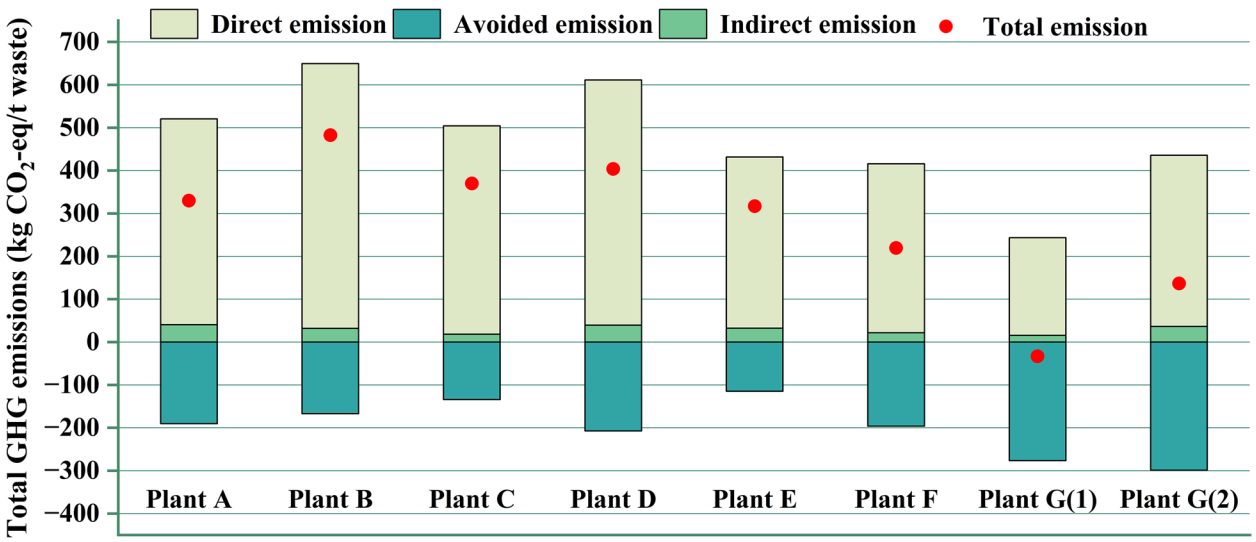


Fig. 7 Total GHG emissions from waste incineration plants

Compared to Method III, there are further reductions in average fossil CO_2 emissions calculated using Methods IV and V based on flue gas analysis. However, it is worth noting that the sampling and characterization of solid waste are more uncertain than those of flue gas samples, which are relatively homogeneous and stable in nature. The use of averages to compare the fossil CO_2 emissions by several methods is not justified, and the uncertainties in the calculated parameters should be fully considered.

Figure 6 illustrates the distribution of fossil CO_2 emission results for incineration plants A–G using the five methods, taking full account of the uncertainties in the calculation parameters (Table S4). The distributions of fossil CO_2 emissions vary considerably in different MSWI plants due to the fluctuating physical composition of waste, but the overall trend is the same, with the range of fossil CO_2 emissions gradually approaching the validation values of Methods IV and V on going from Method I to Method III. This shows that it is reasonable and necessary to improve generic Method I by invoking co-incinerated waste and actual bottom ash production, which is highly conducive to improving the accuracy of accounting for fossil CO_2 from waste incineration.

3.4 Life cycle GHG emissions by the improved method

The total climate change contributions of MSWI plants A–G are given in Fig. 7. Direct emissions are based on the results of fossil CO_2 emission obtained from Method III, indirect emission accounting is based on operational data from MSWI plants, and avoided emissions are based on electricity substitution. Electricity exchange is based on the marginal mixes in the area in which the incineration plant is located, amounting to 0.420 kg CO_2-eq/kWh for plants A–E, 0.695 kg CO_2-eq/kWh for plant F, and 1.092 kg CO_2-eq/kWh for plants G(1) and G(2). The total carbon emissions from all MSWI plants ranged from −33.2 to 483 kg CO_2-eq/t waste, with large discrepancies between different MSWI plants.

In all of the studied MSWI plants, indirect emissions were in the range of 15.59–40.39 kg CO_2-eq/t waste, corresponding to only 3.80%–9.06% of the direct emissions. The main concern when considering carbon emissions from waste incineration remains direct and avoided emissions. The improved method of direct emission accounting shows that it is mainly affected by the physical composition of the waste, especially the content of fossil source waste. Better waste management and recycling of fossil source waste with high recycling value, such as plastics, may yield better carbon emission reduction benefits.

The avoided emissions of MSWI plants A–G ranged from −299 to −115 kg CO_2-eq/t waste. It is worth noting that the GHG emission reduction from electricity generation does not depend entirely on the efficiency of the MSWI plant. The carbon emission factor of the electricity in the area in which the incineration plant is situated is a non-negligible factor. MSWI plant D had the highest electricity generation efficiency, 494 kWh/t waste, but its carbon emission reduction through electricity generation was only 207 kg CO_2-eq/t waste, much lower than that of MSWI plant G(2) (−299 kg CO_2-eq/t waste), which had an electricity generation efficiency of only 274 kWh/t waste. This is due to the fact that the carbon emission factor for electricity in MSWI plant D was 0.420 kg CO_2-eq/kWh, whereas that in MSWI plant G(2) was 2.6 times higher (1.092 kg CO_2-eq/kWh).

The improved Method III reveals a certain reduction in direct carbon emissions from MSWI compared to the generic Method I. Due to the high carbon emission factor, the avoided GHG emission in some areas can offset the direct and indirect GHG emissions to a great extent, and even achieve negative net carbon emissions (−35.1 kg CO_2-eq/t waste in plant G[1]), making waste incineration a carbon sink. However, as the energy mix becomes cleaner, the carbon reduction benefits of waste incineration will diminish, making waste incineration a carbon source. Therefore, reducing direct carbon emissions by increasing the recycling value of fossil waste, and increasing emission reductions from energy utilization by improving the efficiency of electricity generation or developing other energy-efficient thermal energy utilization methods, such as combined heat and power, should be the direction of future efforts to effectively reduce GHG emissions associated with waste incineration.

4 Conclusions

This study proposed an improved method for calculating direct emissions from waste incineration on the basis of the common method (Method I) and their validation by the method of ^{14}C testing of flue gas. The physical composition was adjusted by considering the co-incinerated waste materials (Method II) and the actual bottom ash yield (Method III). Furthermore, life cycle carbon emissions from waste incineration were evaluated by taking into account actual operational data from MSWI plants.

^{14}C testing was applied to obtain the FCFs of different types of solid waste materials that are more in line with current Chinese waste characteristics. The fossil carbon contents of plastics, polyester textiles, and mixed textiles, which play a major role in carbon emissions from waste incineration, showed significant deviations from the IPCC default values, being evaluated as 90.52% ± 0.81%, 93.6% ± 0.02%, and 62.4% ± 0.01%, respectively. Under the eight different incineration conditions, Method III yielded fossil CO_2 emissions of 222–610 kg CO_2-eq/t waste, corresponding to reductions of 3.4–221 kg CO_2-eq/t waste compared to Method I. Flue gas was analyzed to verify Method III. The fossil CO_2 emissions were 249–445 kg CO_2-eq/t waste before flue gas treatment (Method IV) and 233–405 kg CO_2-eq/t waste after flue gas treatment (Method V). After fully considering the uncertainties in the calculation parameters, the results of improved Method III were closer to the validation values than in the case of Method I, and should further improve the accuracy of the calculated results of carbon emissions from waste incineration. Life cycle carbon emissions from MSWI plants, taking into account operational data, ranged from −33.2 to 483 kg CO_2-eq/t waste. That is to say, some incineration plants are currently able to achieve negative net carbon emissions, due to the lower proportion of fossil-source waste materials and the higher local carbon emission factor for electricity.

The results of this study should contribute to more accurate accounting of carbon emissions from waste incineration and provide data and methodological support for GHG emission accounting for waste management. Data of this study on the fossil carbon contents of solid waste samples and waste incineration flue gases derived from ^{14}C tests should greatly enrich the database of waste management and the life cycle inventory of GHG emissions.

Supplementary Information

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Supplementary Material 1.

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Authors' contributions
Study conception, material preparation, data collection and analysis were performed by Yifei Ma and Hua Zhang. Hua Zhang provided the funding acquisition. All writers provided feedback on earlier drafts of the manuscript after Yifei Ma wrote the original draft. All authors have read and approved the final draft.

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Data availability
The datasets used or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests
All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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